16.07 Dynamics — Quiz 3: Bifilar Pendulum (Solutions)

The questions below refer to the T-shaped body and bifilar pendulum from Problem Set 3. Recall that for small oscillations, the period of the bifilar pendulum is

$$P = \frac{2\pi}{d}\sqrt{\frac{J\ell}{mg}},$$

where m is the mass of the body, J is its moment of inertia about the vertical axis through its center of mass, ℓ is the string length, and $2d$ is the string separation.

1. Suppose we measure the moment of inertia of the T-shaped body about a vertical axis that does *not* pass through its center of mass. Will this moment of inertia be smaller than, equal to, or larger than the value about the vertical axis through the center of mass? *Hint:* think about the parallel-axis theorem.

Answer: Larger. By the parallel-axis theorem, the moment of inertia about a parallel axis a distance r from the center of mass is $J' = J + mr^2$. Since $mr^2 > 0$ for any $r \neq 0$, $J' > J$. The axis through the center of mass gives the *smallest* moment of inertia of any family of parallel axes.

2. Suppose we make the T-shaped body from a material with twice the density (so that it's mass doubles), keeping everything else the same. Does its moment of inertia J change? If so, by what factor? Does the period P of the bifilar pendulum change?

Answer: J doubles; P is unchanged. The moment of inertia is $J = \int r^2\, dm = \int r^2 \rho\, dV$. With the shape unchanged, every mass element doubles while its distance r from the axis stays the same, so $J \to 2J$. The period depends on J and m only through the ratio J/m:

$$P = \frac{2\pi}{d}\sqrt{\frac{\ell}{g}\cdot\frac{J}{m}} \;\to\; \frac{2\pi}{d}\sqrt{\frac{\ell}{g}\cdot\frac{2J}{2m}} = P.$$

3. Now suppose we double every dimension of the T-shaped body, but adjust the material density so that its mass m stays the same (and keep everything else the same). By what factor does J change? By what factor does the period P change?

Answer: J increases by a factor of 4; P doubles. The moment of inertia scales as mass times length squared, $J \sim mL^2$. With m fixed and every length doubled, every term grows by 2^2, so $J \to 4J$. Since the strings (d and ℓ) and the mass are unchanged,

$$P \propto \sqrt{\frac{J}{m}} \implies P \to \sqrt{4}\,P = 2P.$$